

Current Flow: Minimal Data Entry for Investments and Pensions

INVESTMENT ACCOUNT (Minimal Data)

Required Fields

- `account_type` (isa, gia, nsi, etc.)
- `provider`
- `current_value`
- `tax_year`

Key Behavior

1. **Auto-created Cash Holding** - When an investment account is created with no holdings, the system automatically creates a 100% Cash holding equal to the account value
(`InvestmentController.php:313-326`)
2. **Returns show 0%** - Because `cost_basis` is null on the cash holding, `calculateReturns()` returns 0%
3. **Diversification score is low** - Single asset type (cash) triggers "Improve Portfolio Diversification" recommendation

Flow

```
AccountForm.vue → investmentService.createAccount() → POST  
/api/investment/accounts  
→ InvestmentController.storeAccount() → InvestmentAccount::create()  
→ Auto-creates Holding (100% Cash)
```

Frontend Form - AccountForm.vue

Location: `resources/js/components/Investment/AccountForm.vue`

Required Fields (Validation):

- `account_type` - Required (must select from: isa, gia, nsi, onshore_bond, offshore_bond, vct, eis, other)
- `provider` - Required (free text, max 255 chars)
- `current_value` - Required (numeric, min 0)
- `tax_year` - Required (defaults to 2025/26)
- `account_type_other` - Required IF `account_type` is 'other'

Optional Fields:

- `platform` - Optional platform/product name
- `country` - Defaults to 'United Kingdom'

- `platform_fee_percent / platform_fee_amount / platform_fee_frequency` - All optional
- `isa_subscription_current_year` - Optional (ISA accounts only)
- `isa_type` - Defaults to 'stocks_and_shares' if not specified
- `ownership_type` - Defaults to 'individual'
- `joint_owner_id, trust_id` - Optional for ownership
- `risk_preference` - Optional (loaded from risk profile if available)

Backend Controller - InvestmentController.php

Location: `app/Http/Controllers/Api/InvestmentController.php` (lines 267-349)

Validation Rules:

```
$validated = $request->validate([
    'account_type' => ['required', Rule::in([...valid types...])],
    'account_type_other' =>
    'required_if:account_type,other|nullable|string|max:255',
    'provider' => 'required|string|max:255',
    'account_number' => 'nullable|string|max:255',
    'platform' => 'nullable|string|max:255',
    'current_value' => 'required|numeric|min:0',
    'contributions_ytd' => 'nullable|numeric|min:0',
    'tax_year' => 'required|string|max:10',
    'platform_fee_percent' => 'nullable|numeric|min:0|max:100',
    'isa_type' => ['nullable', Rule::in([...])],
    'isa_subscription_current_year' => 'nullable|numeric|min:0|max:20000',
    'ownership_type' => ['nullable', Rule::in(['individual', 'joint',
    'trust'])],
    'ownership_percentage' => 'nullable|numeric|min:0|max:100',
    'joint_owner_id' => 'nullable|exists:users,id',
    'trust_id' => 'nullable|exists:trusts,id',
]);
```

Auto-Created Cash Holding (Lines 313-326)

When an investment account is created with minimal data:

```
Holding::create([
    'holdable_id' => $account->id,
    'holdable_type' => InvestmentAccount::class,
    'asset_type' => 'cash',
    'security_name' => 'Cash',
    'allocation_percent' => 100.00, // 100% cash initially
    'current_value' => $account->current_value, // Equals account value
    'quantity' => null, // No quantity data
    'purchase_price' => null, // No price data
    'purchase_date' => null, // No purchase date
    'current_price' => null, // No price data
    'cost_basis' => null, // Cannot calculate without price data
```

```
'ocf_percent' => 0.00, // No fees
]);
```

Calculations with No Holdings Data

Portfolio Analysis - PortfolioAnalyzer.php (lines 24-51):

```
public function calculateReturns(Collection $holdings): array
{
    $totalCostBasis = $holdings->sum('cost_basis'); // Sum of null values
    = 0
    $totalCurrentValue = $holdings->sum('current_value');

    if ($totalCostBasis == 0) {
        return [
            'total_cost_basis' => 0.0,
            'total_current_value' => 0.0,
            'total_gain' => 0.0,
            'total_return_percent' => 0.0, // Returns are 0%
            'ytd_return' => 0.0,
            'one_year_return' => 0.0,
        ];
    }
}
```

Asset Allocation:

- Returns 100% Cash allocation
- Single asset type = low diversification score
- Triggers recommendation: "Improve Portfolio Diversification"

DC PENSION (Minimal Data)

Required Fields (Frontend)

- **pension_type** (occupational, sipp, personal, stakeholder)
- **scheme_name**
- **current_fund_value**
- **annual_salary** (workplace pensions only)

Key Behavior

1. **NO auto-created holdings** - Unlike investments, DC pensions do NOT create a default cash holding
2. **Projections use current value only** - With no monthly contributions, projection = $\text{current_value} * (1 + \text{growth_rate})^{\text{years}}$
3. **Diversification returns "no holdings"** - GET `/api/retirement/pensions/dc/{id}/diversification` returns `has_holdings: false`

Flow

```
DCPensionForm.vue → retirementService.createDCPension() → POST  
/api/retirement/pensions/dc  
→ RetirementController.storeDCPension() → DCPension::create()  
→ No auto-created holdings
```

Frontend Form - DCPensionForm.vue

Location: `resources/js/components/Retirement/DCPensionForm.vue`

Required Fields (Validation - lines 520-561):

- `pension_type` - Required (occupational, sipp, personal, stakeholder)
- `scheme_name` - Required (free text)
- `current_fund_value` - Required (numeric, min 0)
- `annual_salary` - Required for workplace pensions only

Optional Fields:

- `provider` - Optional (free text)
- `policy_number` - Optional (free text)
- `expected_return_percent` - Defaults to 5.0 if not provided
- `platform_fee_percent` - Optional
- `lump_sum_contribution` - Optional (SIPP/personal only)
- `salary_sacrifice` - Checkbox (workplace only), defaults to false
- `retirement_age` - Optional (55-75)
- `notes` - Optional (textarea)
- `risk_preference` - Optional

Backend Validation - StoreDCPensionRequest.php

Location: `app/Http/Requests/Retirement/StoreDCPensionRequest.php`

All Fields Are Nullable:

```
public function rules(): array  
{  
    return [  
        'scheme_name' => ['nullable', 'string', 'max:255'],  
        'scheme_type' => ['nullable', 'in:workplace,sipp,personal'],  
        'pension_type' => ['nullable',  
        'in:occupational,sipp,personal,stakeholder'],  
        'provider' => ['nullable', 'string', 'max:255'],  
        'member_number' => ['nullable', 'string', 'max:255'],  
        'current_fund_value' => ['nullable', 'numeric', 'min:0'],  
        'annual_salary' => ['nullable', 'numeric', 'min:0'],  
        'employee_contribution_percent' => ['nullable', 'numeric',  
        'min:0', 'max:100'],
```

```

        'employer_contribution_percent' => ['nullable', 'numeric',
'min:0', 'max:100'],
        'monthly_contribution_amount' => ['nullable', 'numeric', 'min:0'],
        'lump_sum_contribution' => ['nullable', 'numeric', 'min:0'],
        'investment_strategy' => ['nullable', 'string', 'max:255'],
        'platform_fee_percent' => ['nullable', 'numeric', 'min:0',
'max:10'],
        'retirement_age' => ['nullable', 'integer', 'min:55', 'max:75'],
        'projected_value_at_retirement' => ['nullable', 'numeric',
'min:0'],
    ];
}

```

Database Storage - DCPension Model

Location: `app/Models/DCPension.php`

Fields Stored (with minimal data):

```

[
    'user_id' => 1,
    'scheme_name' => 'My Personal Pension',
    'scheme_type' => 'personal',
    'pension_type' => 'personal',
    'provider' => 'Aviva',
    'member_number' => null,
    'current_fund_value' => 75000,
    'annual_salary' => null,
    'employee_contribution_percent' => null,
    'employer_contribution_percent' => null,
    'monthly_contribution_amount' => null,
    'lump_sum_contribution' => null,
    'investment_strategy' => null,
    'platform_fee_percent' => null,
    'retirement_age' => null,
    'expected_return_percent' => 5,
    'projected_value_at_retirement' => null,
    'has_custom_risk' => false,
    'risk_preference' => null,
]

```

Pension Projections with Minimal Data

PensionProjector.projectDCPension() (lines 36-58):

```

public function projectDCPension(DCPension $pension, int
$yearsToRetirement, float $growthRate): float
{
    $currentValue = 75000; // Provided

```

```
$monthlyContribution = null ?? 0.0; // Defaults to 0
$annualContribution = 0;

$netGrowthRate = 0.05 - (0 / 100); // 5% - 0% fees = 5%

// Future value of current fund only (no contributions)
$futureValueOfCurrentFund = 75000 * pow(1.05, $yearsToRetirement);

return $futureValueOfCurrentFund;
}
```

Example: 10 years to retirement

- Current: £75,000
- Monthly Contribution: £0
- Growth Rate: 5% p.a.
- Projected at Retirement: $£75,000 * 1.05^{10} = £122,390$

Diversification Analysis with No Holdings

RetirementController.getDCPensionDiversification() (lines 570-622):

```
if ($holdings->isEmpty()) {
    return response()->json([
        'success' => true,
        'data' => [
            'message' => 'No holdings recorded for this pension',
            'has_holdings' => false,
            'pension_id' => $id,
            'pension_name' => 'My Personal Pension',
        ],
    ], 1);
}
```

Key Differences Summary

Aspect	Investment Account	DC Pension
Auto-Created Holding	YES - 100% Cash	NO
Required Fields	account_type, provider, current_value, tax_year	scheme_name, pension_type, current_fund_value
Fees Default	0% (nullable)	0% or null (nullable)
Holdings Required	No (but auto-created)	No

Aspect	Investment Account	DC Pension
Projections Without Data	Returns 0% (cost_basis is null)	Growth of current_value only (no contributions)
Risk Profile	Can have account-level override	Can have pension-level override
Joint Ownership	Supported (single-record pattern)	Not applicable (pensions are personal)
Diversification Analysis	Returns score + recommendations	Returns "no holdings" message if empty

Display & User Feedback

For Investment Accounts (Minimal Data):

- Dashboard shows account value: £50,000
- Asset allocation: 100% Cash
- Returns: 0% (no cost basis data to calculate)
- Recommendation triggered: "Improve Portfolio Diversification"
- User prompted to add holdings

For DC Pensions (Minimal Data):

- Dashboard shows current fund: £75,000
- Projected at retirement (10 years, 5% p.a.): £122,390
- Monthly contributions: £0
- No diversification analysis (no holdings)
- Recommendation may trigger: "Increase Pension Contributions" (if income gap exists)

Strategy Flow & Decision Trees

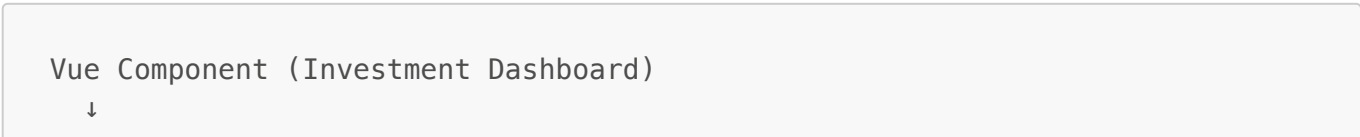
Investment Account Strategies

Strategy Architecture

Core Components:

- **InvestmentAgent** ([app/Agents/InvestmentAgent.php](#)) - Orchestrates portfolio analysis and generates basic recommendations
- **PortfolioStrategyService** ([app/Services/Investment/PortfolioStrategyService.php](#)) - Aggregates all investment strategies with priority ordering
- **InvestmentRecommendation** model - Stores strategy data for tracking

Request Flow:



```
PortfolioStrategyController API
↓
PortfolioStrategyService (aggregates all strategies)
↓
TaxOptimizationAnalyzer, FeeAnalyzer, DriftAnalyzer
↓
InvestmentAgent (fallback simple analysis)
↓
Frontend Display (RecommendationsSection.vue,
StrategyRecommendationCard.vue)
```

Strategy Categories & Decision Triggers

The system generates recommendations in a **4-level priority hierarchy**:

PRIORITY 1: TAX STRATEGIES (Highest Priority)

Decision Logic: `PortfolioStrategyService` (line 85-92)

- Calls `TaxOptimizationAnalyzer::analyzeCompleteTaxPosition()`

Strategies:

- **ISA Underutilization** - When ISA allowance unused (£20,000 annual)
- **Bed & ISA** - Selling positions at loss then rebuy in ISA
- **Tax Loss Harvesting** - When unrealized losses exist to offset gains

Thresholds:

- Tax efficiency score < 80/100 triggers general optimization recommendation
- Harvesting opportunities count > 0 with potential savings calculated

Potential Savings:

- ISA: up to 20% tax savings on gains for basic rate, 40% for higher rate
- Bed & ISA: full tax deferral on reinvested amount

PRIORITY 2: WRAPPER STRATEGIES (Bond Wrappers)

Decision Logic: `PortfolioStrategyService` (line 94-98, 187-257)

- Only shows for GIA/General Investment Accounts (not ISA)
- Only for **higher and additional rate taxpayers**

Eligibility Thresholds:

```
BOND_WRAPPER_MIN_BALANCE = £50,000
OFFSHORE_BOND_MIN_BALANCE = £100,000 (additional rate taxpayers only)
```

Trigger Conditions:

1. User is higher/additional rate taxpayer
2. GIA account balance $\geq$ £50,000
3. For offshore: balance $\geq$ £100,000 AND additional rate taxpayer

Calculation of Benefit:

Estimated Return: 6% annual
 Annual Growth: Account Value $\times$ 6%
 Tax Deferral Benefit:
 - Additional rate: 25% of growth (45% income tax vs ~20% effective bond rate)
 - Higher rate: 20% of growth (40% income tax vs ~20% effective bond rate)

Recommendation Selection:

- Balance $\geq$ £100,000 AND additional rate $\rightarrow$ Offshore Bond (recommended)
- Balance $\geq$ £50,000 AND higher/additional rate $\rightarrow$ Onshore Bond

PRIORITY 3: FEE STRATEGIES

Decision Logic: [PortfolioStrategyService](#) (line 100-107)

- Uses FeeAnalyzer to assess portfolio costs

Trigger #1: Portfolio-Level High Fees

- Current fee percentage $>$ industry average
- Levels: "high" (5-7% annually) or "very_high" ($>7\%$)
- Shows when: Assessment level = 'high' or 'very_high'

Trigger #2: Individual High-Fee Holdings

- Individual fund OCF $>$ 0.5% annually
- Minimum threshold: Potential savings $>$ £50/year

Calculation:

Current Annual Cost: Holding Value $\times$ Current OCF
 Low-Cost Alternative: Holding Value $\times$ 0.15% (assumed)
 Potential Saving: $\max(0, \text{Current Cost} - \text{Low-Cost Cost})$

PRIORITY 4: REBALANCING STRATEGIES

Decision Logic: [PortfolioStrategyService](#) (line 109-113)

- Uses DriftAnalyzer to evaluate allocation deviation

Trigger Conditions:

```
if (urgency.action_required === true) {
  // Show rebalancing recommendation
}
```

Drift Analysis:

- Compares current allocation vs. target allocation from risk profile
- Calculates drift percentage per asset class
- Default threshold: 10% drift from target

Urgency Levels:

- **High:** Drift > 15% from target
- **Medium:** Drift 10-15% from target
- **Low:** Drift < 10% from target

Investment Decision Tree

```
User has investment accounts?
├─ NO → Return empty
└─ YES
    ├─ PRIORITY 1 (Tax): Tax efficiency score < 80?
    │   ├── YES → Tax strategy recommendations
    │   └─ Harvesting opportunities? → Tax loss harvest rec
    ├─ PRIORITY 2 (Wrapper): Higher/additional rate taxpayer?
    │   ├── YES + GIA balance >= £50,000
    │   │   └─ Recommend onshore/offshore bond wrapper
    │   └─ NO → Skip
    ├─ PRIORITY 3 (Fees): Portfolio fees > industry average?
    │   ├── YES → High fee portfolio recommendation
    │   └─ Any holdings with OCF > 0.5%?
    │       ├── YES + savings > £50 → High fee holding recs
    │       └─ NO → Skip
    └─ PRIORITY 4 (Rebalancing): Drift > threshold?
        ├── YES → Rebalancing recommendation
        └─ NO → Skip
```

DC Pension Strategies (Retirement Planning)

Strategy Architecture

Core Components:

- **RetirementStrategyService**
(`app/Services/Retirement/RetirementStrategyService.php`) - Main strategy engine
- **ContributionOptimizer** (`app/Services/Retirement/ContributionOptimizer.php`) - Contribution analysis
- **RetirementAgent** (`app/Agents/RetirementAgent.php`) - High-level orchestration
- **StrategyCard.vue** - Frontend display with interactive sliders

Request Flow:

```
Vue Component (Retirement Dashboard)
↓
RetirementStrategyController API
↓
RetirementStrategyService::getStrategies()
↓
RetirementProjectionService (gets base projections)
↓
Four Priority Checks (in sequence)
↓
StrategyCard.vue with Interactive Sliders
```

Probability-Based Decision System

"On Track" Threshold: 95% probability

Current position starts with a calculated probability based on:

- Current retirement income projection
- Target retirement income
- Years to retirement
- Current pension pot value
- Guaranteed income (DB pensions, state pension)

Probability Formula:

```
probability = 10 + (income_ratio × 85)
where income_ratio = projected_income / target_income

// Capped at 10% (very poor) to 95% (on track)
```

Strategy Priorities (Sequential Evaluation)

PRIORITY 1: EMPLOYER MATCH OPTIMIZATION

Trigger Conditions: (`RetirementStrategyService` line 424-500)

1. Scheme Type Check:

- Only applies to workplace pensions
- Skips personal pensions (SIPP), stakeholder, etc.

2. Match Detection:

- Uses robust inference algorithm (line 514-542)
- Checks explicit `employer_matching_limit` field (if valid 1-50%)
- Falls back to inferring from contribution patterns:
 - If `employer_contribution > employee_contribution` → infer match limit at employer level
 - If `employee ≥ employer` → no matching opportunity
 - If `employer ≤ 0` → no matching scheme

3. Gap Detection:

- Current employee contribution < inferred match limit
- AND gap ≥ 0.5% (meaningful difference)

Calculation:

```
Additional Percentage Needed = Match Limit - Current Employee %
Additional Monthly = (Annual Salary × Additional %) / 12
Employer Bonus = Employee Additional (1:1 match)
Total Additional Monthly = Employee + Employer
```

Impact on Probability:

- Each £100/month additional contribution ≈ +2% probability
- Shows with cumulative tracking from prior strategies

PRIORITY 2: CONTRIBUTION INCREASE STRATEGY

Trigger Conditions: (`RetirementStrategyService` line 546-631)

1. Affordability Check:

- Disposable income > 0 (from income & expenditure profile)
- AND remaining annual allowance > 0

2. Constraint Analysis:

- Max additional = min(disposable_income, remaining_allowance)
- Recommended = 50% of max additional (conservative suggestion)
- Respects annual allowance carry-forward (if 3-year history exists)

3. Binding Constraint Determination:

- If `annual_allowance < disposable_income` → allowance is limiting
- Otherwise → affordability is limiting

Slider Configuration:

```
'slider_config' => [
  'min' => current_monthly_total,
  'max' => current_monthly + max_additional,
  'step' => £50,
  'unit' => '/month',
  'format' => 'currency',
]
```

Impact Calculation:

```
Additional Monthly (slider input - current)
↓
Convert to annual: × 12
↓
Project to retirement: FV of annuity formula with 5% growth
↓
Convert to annual income: pot × 0.047 (sustainable withdrawal)
↓
Recalculate probability with cumulative prior strategies
```

Only Shows If:

- Still not on track after employer match strategy
- Probability < 95%

PRIORITY 3: RETIREMENT AGE ADJUSTMENT

Trigger Conditions: ([RetirementStrategyService](#) line 636-752)

1. Age Constraint Check:

- Current retirement age < 68 (max recommended)
- Slider max = 75

2. Income Target Requirement:

- Target income > 0
- (Cannot calculate without target)

3. Gap Analysis:

- Income gap > £5,000 → suggests retirement age adjustment

Optimal Age Calculation:

```
// Test each year of delay
for ($yearsDelay = 1 to $maxRecommended) {
  pot_at_delayed_age = calculate_with_monte_carlo()
```

```

sustainable_income = pot × 0.047
total_income = sustainable + guaranteed
coverage = total / target

if (coverage >= 95%) {
    use this age (minimum delay to reach 95%)
    break;
}

// If still can't reach 95%, use max recommended age (68)

```

Cumulative Projection:

"Without Strategy" = Monte Carlo baseline + prior strategies' pot
 "With Strategy" = Pot continues growing for additional years
 Shows: additional pot growth, additional contributions, total at delayed retirement

PRIORITY 4: INCOME TARGET REDUCTION STRATEGY

Trigger Conditions: ([RetirementStrategyService](#) line 759-817)

1. Alternative to Contributions:

- Only if still not on track after prior 3 strategies
- Probability < 95%

2. Realistic Check:

- Total achievable income >= 95% of current target
- (No point if user is way off target)

3. Calculation:

- Recommended target = total achievable income
- (Shows: "accept what you'll actually get, you'll be on track")

Income Components Visible:

Guaranteed Income (DB pensions + state pension)
 + Sustainable from DC pot (pot × 0.047)
 = Total Achievable Income

Only Shows If:

- current_achievable >= 95% of target (realistic path to on-track)
- Probability improves materially by accepting lower target

Cumulative Strategy Tracking

Problem Being Solved:

- Each strategy shows impact considering ALL previous strategies
- Prevents double-counting of contributions
- Shows true cumulative probability improvement

Implementation:

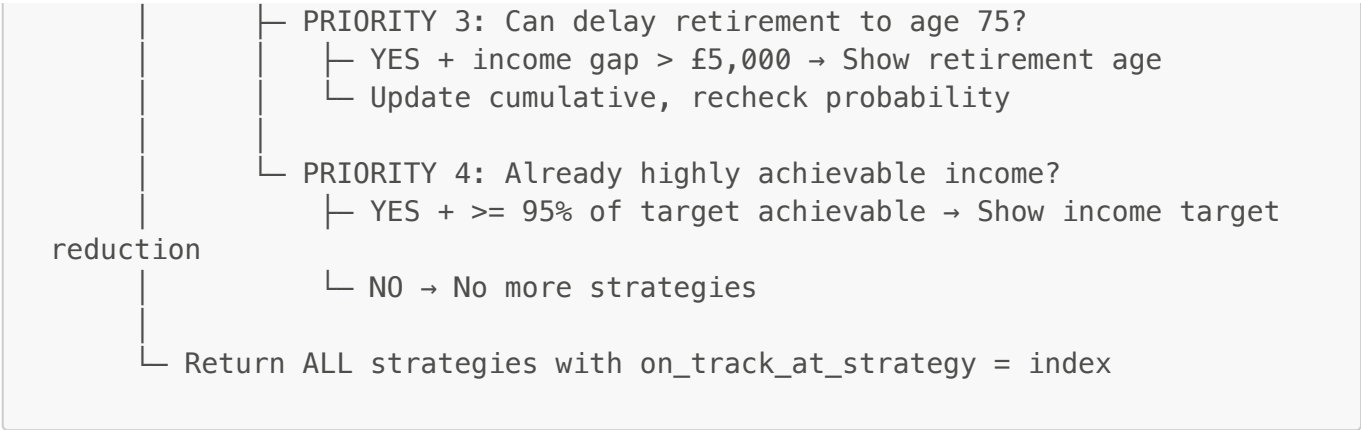
- Each strategy receives:
 - **prior_cumulative_monthly** - total additional monthly from all prior strategies
 - **prior_cumulative_income** - total additional annual income from all prior
 - **prior_probability** - probability AFTER prior strategies (becomes new baseline)
- Strategies store:
 - **prior_probability** - baseline before this strategy
 - **new_probability** - after this strategy
 - **probability_improvement** - delta (only for this strategy)

Example Flow:

```
Initial Probability: 65%
↓
+ Employer Match (adds £200/month) → 75% (+10% improvement)
↓
+ Contribution Increase (adds £300/month, shown on £200 baseline) → 82%
(+7% improvement)
↓
+ Retirement Age Delay → 95% (+13% improvement) → ON TRACK!
```

Pension Decision Tree

```
User has retirement profile?
├─ NO → Return empty
└─ YES
    ├─ Calculate current probability from projections
    └─ Is probability >= 95%?
        ├─ YES → "On track" - no strategies shown
        └─ NO
            ├─ PRIORITY 1: Workplace pension with unmatched employer?
            │   ├─ YES + gap >= 0.5% → Show employer match strategy
            │   └─ Update cumulative, recheck probability
            └─ PRIORITY 2: Disposable income > 0?
                ├─ YES + allowance remaining → Show contribution increase
                └─ Update cumulative, recheck probability
```



Thresholds & Configuration

Investment Thresholds

Threshold	Value	Logic
Tax Efficiency Score	< 80/100	Room for tax optimization
OCF (High Fee)	> 0.5%	Actively managed vs. index
Fee Minimum Saving	> £50/year	Trivial savings ignored
Bond Wrapper Min	£50,000	Transaction cost justified
Offshore Bond Min	£100,000	Costs for additional features justified
Rebalancing Drift	10% (default)	Customizable per account

Pension Thresholds

Threshold	Value	Logic
On Track Probability	>= 95%	High confidence retirement ready
Employer Match Gap	>= 0.5%	Meaningful difference
Income Gap Trigger	> £5,000	Material shortfall
Growth Rate (projection)	5%	Conservative assumption
Withdrawal Rate	4.7%	Sustainable drawdown
Employer Match Limit	1-50%	Reasonable salary % range
Annual Allowance	£60,000	2025/26 UK limit

Frontend Components

Investment Strategy Components

- **Recommendations.vue** - Main page with load button
- **InvestmentRecommendationsTracker.vue** - Shows all strategies

- **StrategyRecommendationCard.vue** - Individual card display
- **TaxStrategySection.vue** - Tax optimization details
- Modals: ISATransferModal, BondWrapperInfoModal, HarvestLossModal, BedAndISAWizardModal

Retirement Strategy Components

- **StrategyCard.vue** - Interactive strategy with slider
 - Real-time probability calculation on slider input
 - Projection chart showing pot growth with/without strategy
 - Income comparison: current path vs. strategy path
 - Cumulative impact tracking
-

API Endpoints

Investment

- **GET /api/investment** - Portfolio strategy analysis
- **POST /api/portfolio-strategy** - Generate strategies

Retirement

- **GET /api/retirement** - Full retirement analysis
- **POST /api/retirement/strategies** - Generate strategies
- **POST /api/retirement/calculate-strategy-impact** - Slider calculation

Aggregation

- **GET /api/recommendations** - All module recommendations (via RecommendationsAggregatorService)